Maharashtra Education Society's
Institute of Management and Career Courses (IMCC), Pune
AUTONOMOUS

[CGPA-3.32] in Cycle 1

Academic Year 2026-27
FY MCA- SEM I
Python
Class Assignment

Roll No: 2601172
Name: Samarth U. Sawant

Q1. Write a program to read a file and count how many vowels are present in its content

```
4   f = open("text.txt", "r")
5
6   data = f.read()
7   count = 0
8
9   for ch in data:
10      if ch.lower() in "aeiou":
11          count = count + 1
12
13  print("Total vowels:", count)
14
15  f.close()
```

Output:

```
PS E:\IMCC\Python\Assignment_1> py Q1.py
Total vowels: 128
PS E:\IMCC\Python\Assignment_1>
```

Q2. Read all lines from a file into a list. Print the total number of lines and display the 2nd and 4th line.

```
f = open("text.txt", "r")

lines = f.readlines()

print("Total number of lines:", len(lines))
print("2nd line:", lines[0], end="")

f.close()
```

Output:

```
PS E:\IMCC\Python\Assignment_1> py Q1.py
Total vowels: 128
PS E:\IMCC\Python\Assignment_1> py Q2.py
Total number of lines: 6
2nd line: The old clock on the mantelpiece
PS E:\IMCC\Python\Assignment_1>
```

Q3 Insert node at a specific position

```
5   sentence = input("Enter a sentence: ")
6
7   words = sentence.split()
8
9   longest = max(words, key=len)
10
11  print("Longest word:", longest)
```

Output:

```
PS E:\IMCC\Python\Assignment_1> py Q3.py
Enter a sentence: It contains a subject and a verb and can stand alone as a complete sentence.
Longest word: sentence.
PS E:\IMCC\Python\Assignment_1>
```

Q4: Find Middle node and print its value

Read a file, split its content into words, and store them in a list
i) Print the total number of words
ii)Print all unique words

```
file = open("text.txt", "r")

content = file.read()
words = content.split()

print("Total number of words:", len(words))

unique_words = set(words)
print("Unique words:", unique_words)

file.close()
```

Output:

```
PS E:\IMCC\Python\Assignment_1> py Q4.py
Total number of words: 77
Unique words: {'rather', 'slowly', 'of', 'fingers,', 'from', 'ticked', 'light', 'black', 'set', 'a', 'in', 'across', 'sink', '
roll', 'tea', 'autumn', 'cold', 'breeze', 'seconds.', 'mantelpiece', 'grey', 'deliberate', 'leaves', 'tree,', 'edge', 'fog', '
into', 'path.', 'harbor.', 'warmth', 'She', 'The', 'her', 'mug,', 'rhythm', 'measure', 'time', 'yellow', 'on', 'them', 'cerami
c', 'Outside', 'clock', 'watched', 'with', 'cobblestone', 'down', 'to', 'rustled', 'letting', 'scattering', 'damp', 'old', 're
maining', 'seemed', 'than', 'heartbeats', 'window,', 'slow,', 'and', 'that', 'the', 'oak'}
PS E:\IMCC\Python\Assignment_1>
```